A Global Perspective on the Past, Present, and Future of Video Streaming Over Starlink

Liz Izhikevich
Reese Enghardt, Te-Yuan Huang, Renata Teixeira





Millions of users rely on Starlink

Starlink @Starlink · 2d

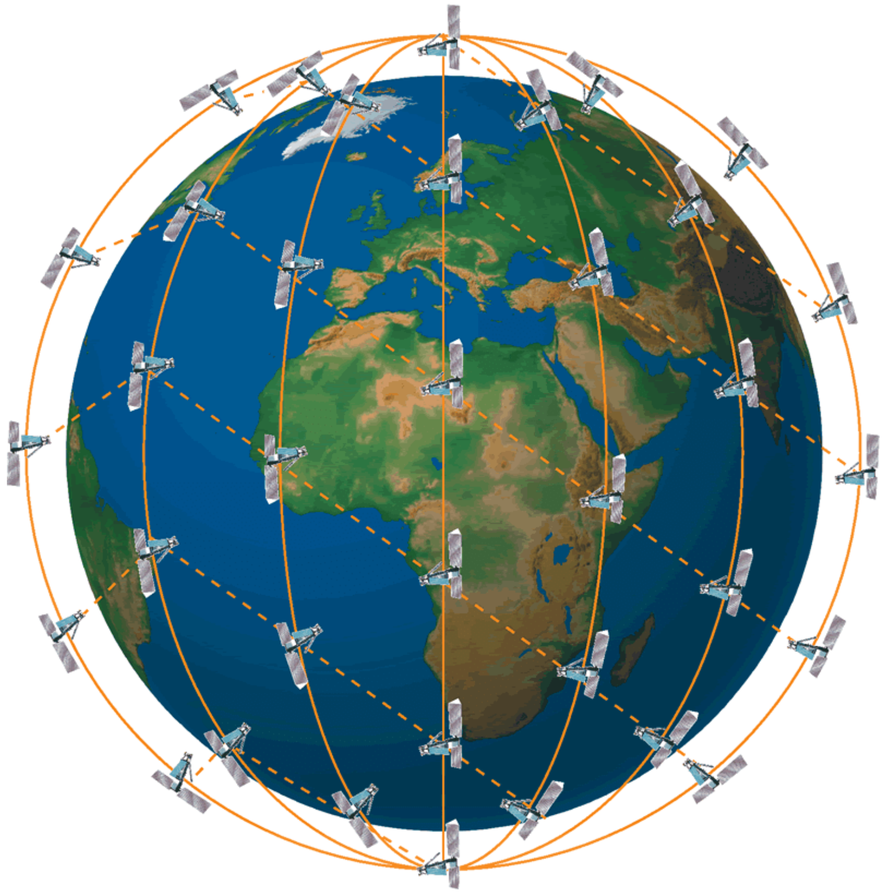
Starlink is connecting more than 6M people with high-speed internet across 140 countries, territories and many other markets.

Thank you to all our customers around the world! 🛰️🌎❤️ → starlink.com/stories-6M



707 1.2K 6.9K 18M

Starlink is a Low Earth Orbit (LEO) Satellite Network



- LEO Satellites work in constellations (e.g., 100s-1000s satellites)
    - High global coverage
- LEO satellites orbit 300km -2000km from Earth
    - Low Latency: minimum RTT (~10ms), bounded by the speed of light



Internet latency standards < 100ms

Sağ, Emirhan & Kavas, Aktül. (2018). Modelling and Performance Analysis of 2.5 Gbps Inter-satellite Optical Wireless Communication (IsOWC) System in LEO Constellation. Journal of Communications. 13. 553-558. 10.12720/ jcm.13.10.553-558.

How does LEO affect video streaming?





- What is **LEO’s role** in video delivery?

Video streaming over LEO is rapidly rising in popularity, especially in Africa



- What is **LEO's role** in video delivery?

- What is the **quality of experience** when streaming video over LEO?



- What is **LEO's role** in video delivery?

- What is the **quality of experience** when streaming video over LEO?

Often equivalent or better than terrestrial connections…but increased bitrate switches and rebuffers



- What is **LEO's role** in video delivery?

- What is the **quality of experience** when streaming video over LEO?

- How can we improve **bitrate switches** and **network rebuffers** for video streaming over LEO?



- What is **LEO's role** in video delivery?

- What is the **quality of experience** when streaming video over LEO?

- How can we improve **bitrate switches** and **network rebuffers** for video streaming over LEO?

Existing **congestion control** & **adaptive bitrate** design principles can partially help, but are not sufficient

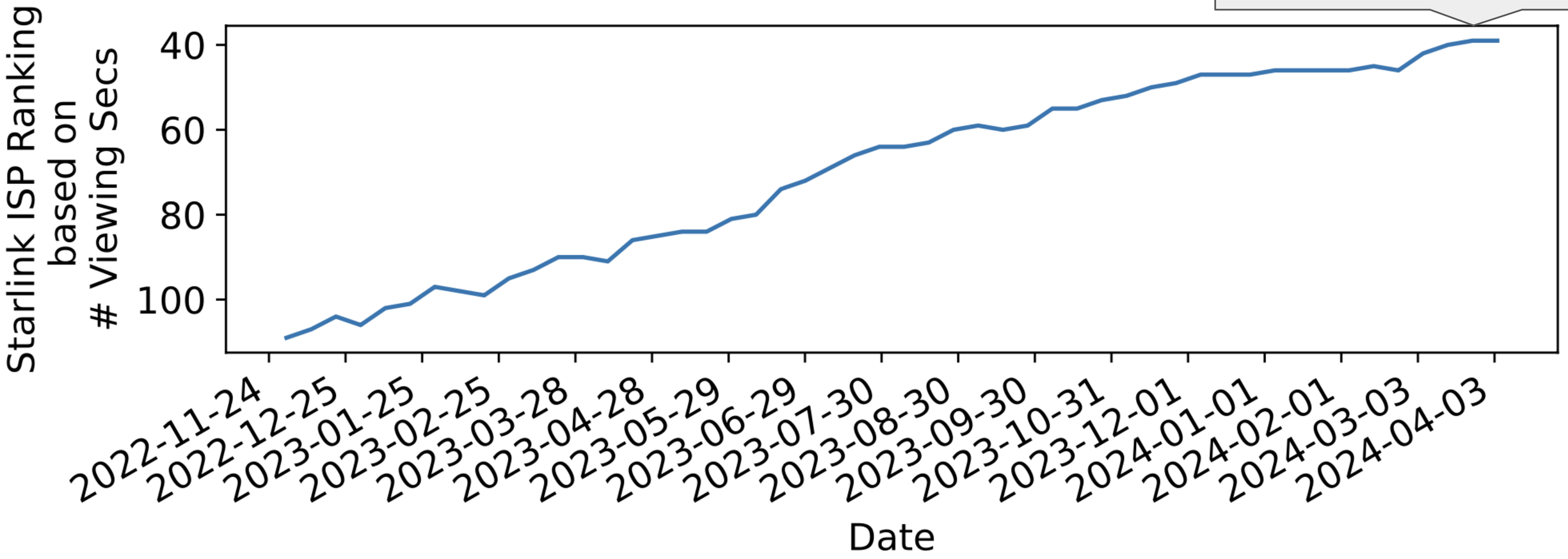


What is LEO’s role in video delivery?

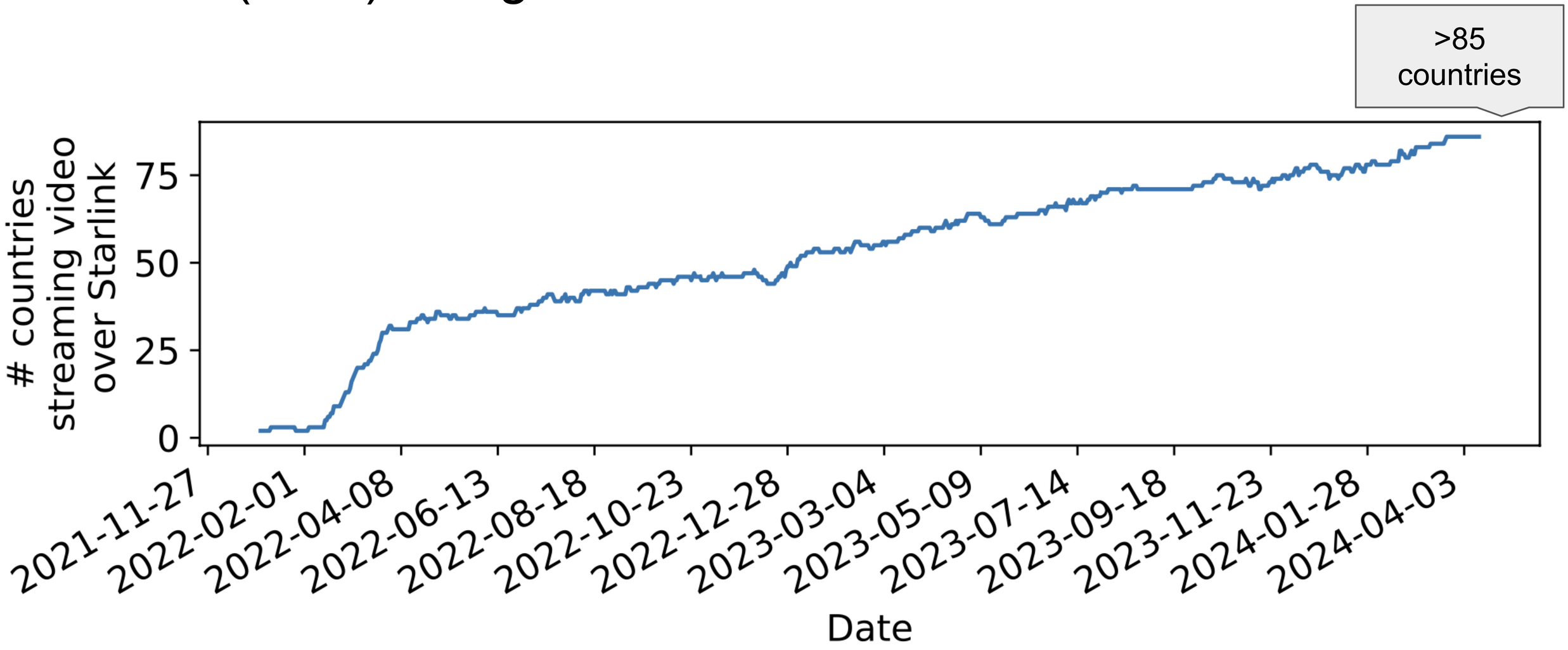
Starlink is the only major LEO provider from which Netflix users stream from today

Starlink (LEO) is growing in popularity

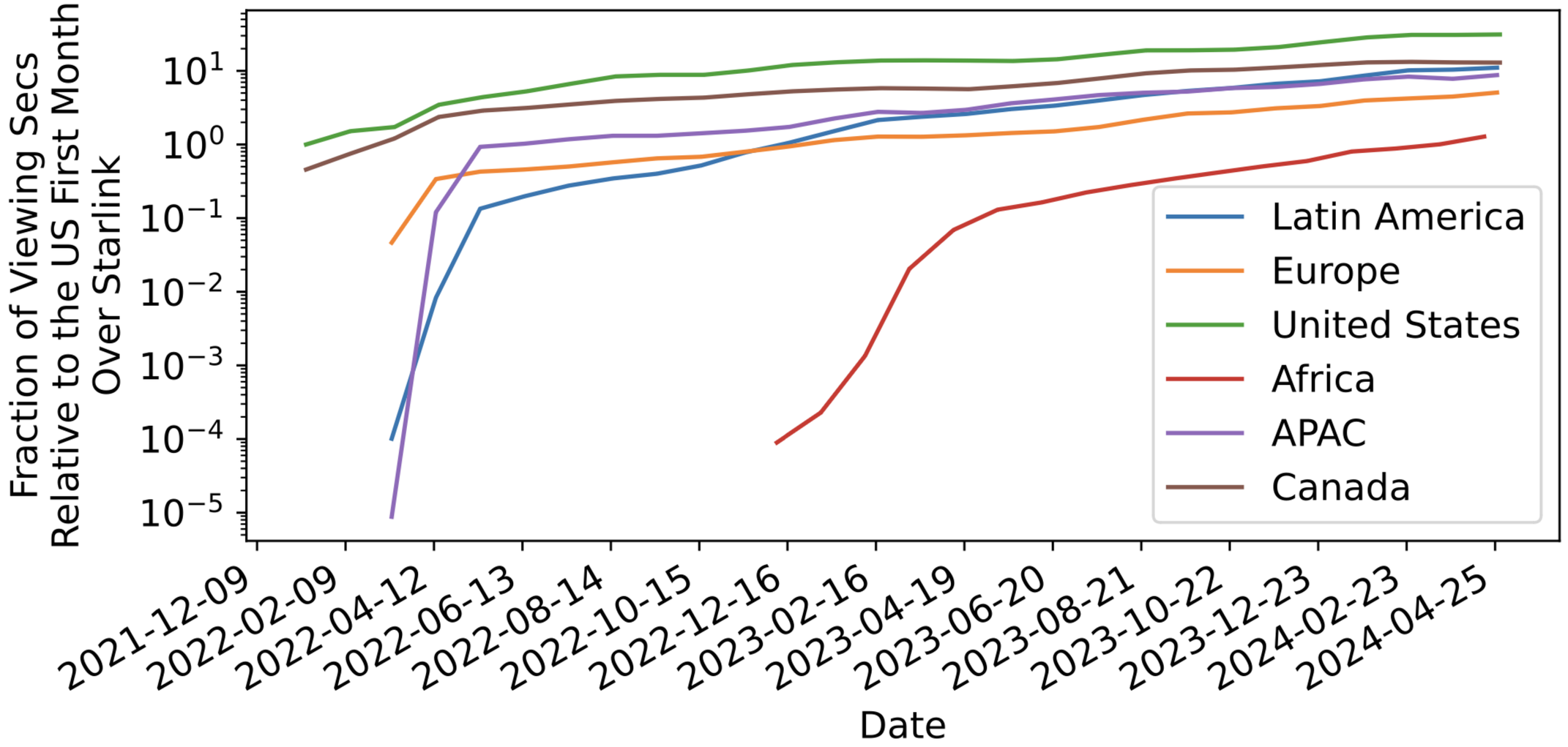
Starlink is 38/20K most popular ISP



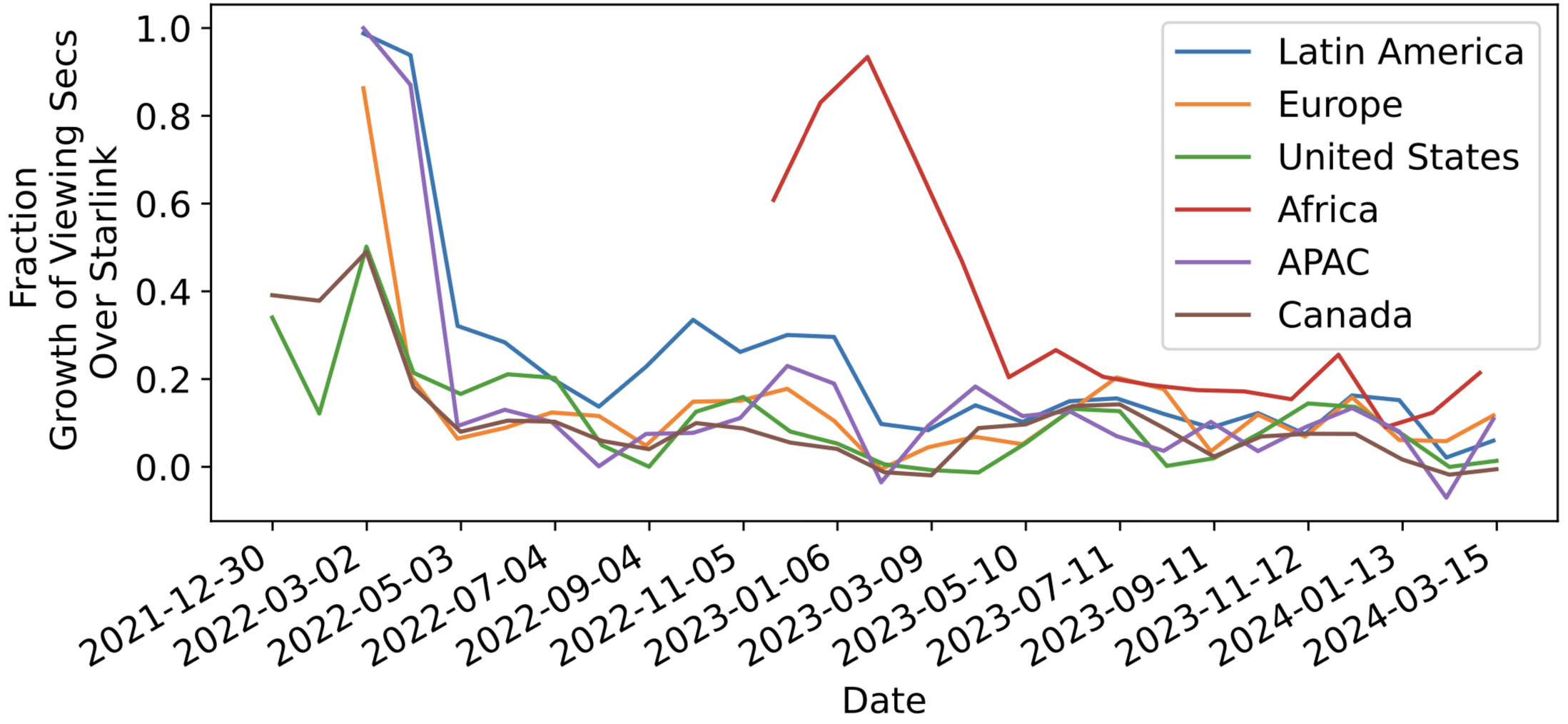
Starlink (LEO) is a *global* ISP



The US streams the most over Starlink



Africa is the fastest growing region for video streaming over Starlink

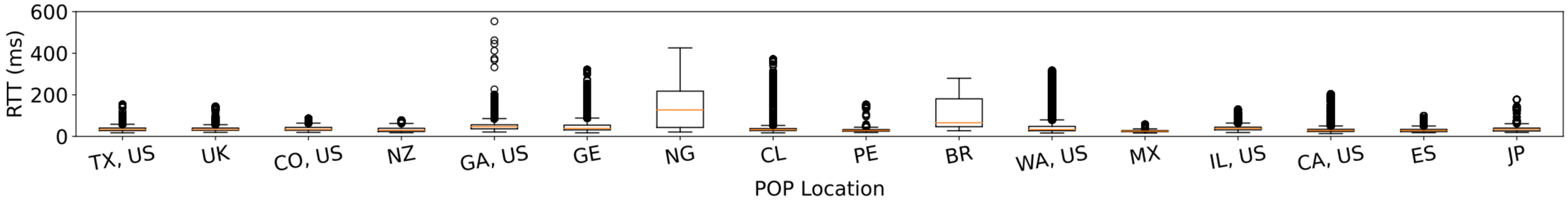


What is the QoE when streaming over Starlink?





Customers experience unexpected latency, depending upon geographic location

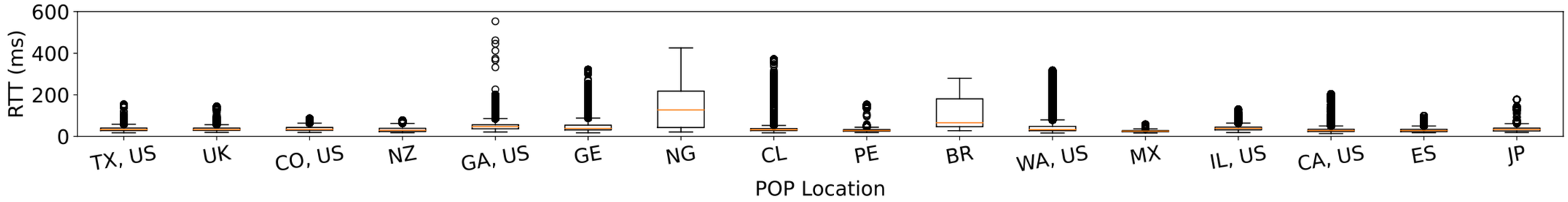


Democratizing LEO Satellite Network Measurement

Authors: Liz Izhikevich, Manda Tran, Katherine Izhikevich, Gautam Akiwate, Zakir Durumeric | Authors Info & Claims

SIGMETRICS/PERFORMANCE '24: Abstracts of the 2024 ACM SIGMETRICS/IFIP PERFORMANCE Joint International Conference on Measurement and Modeling of Computer Systems
Pages 15 - 16 • https://doi.org/10.1145/3652963.3655052

Customers experience unexpected latency, depending upon geographic location



Open Questions:

- How do Starlink latencies compare to regional network alternatives?
- How does latency impact real applications…like video streaming?

What is the QoE when streaming over Starlink?

- Overall Perceptual Video Quality
- Bitrate Switches
- Network Rebuffers

More metrics in paper

Filtering Methodology

We filter for video streaming sessions that are

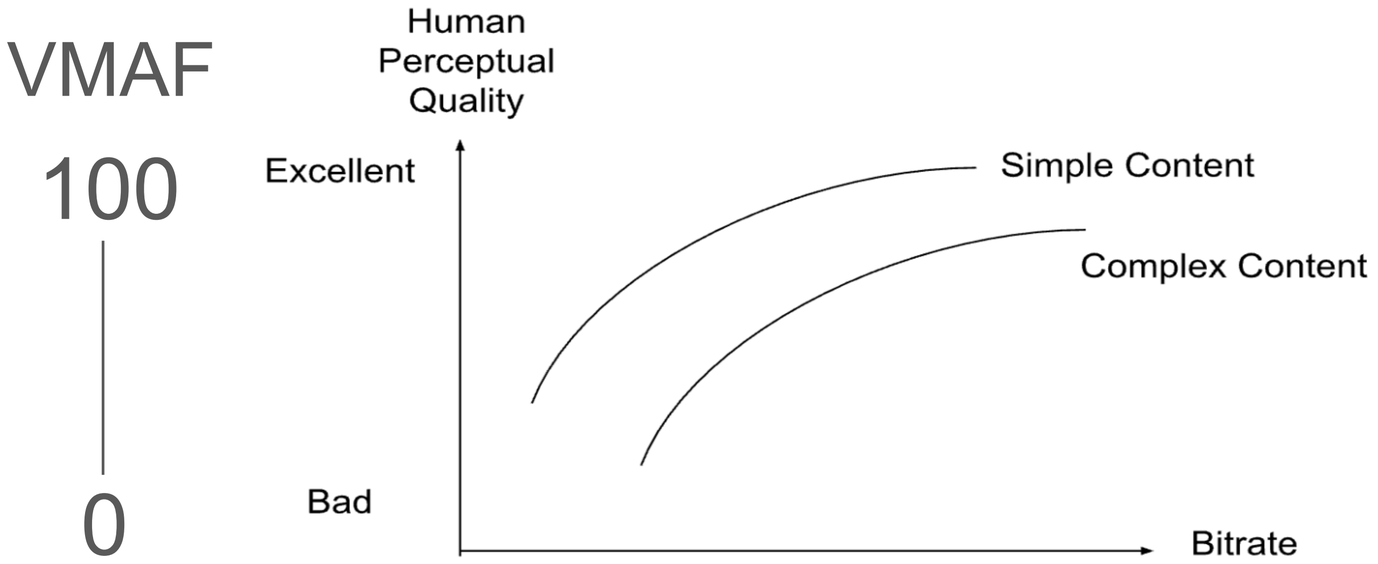
(1) theoretically capable to stream at least at a 720p high definition
(2) at least 5 minutes long
(3) destined towards TVs
(4) streamed during the first week of April 2024

Millions of streaming sessions for Starlink alone!

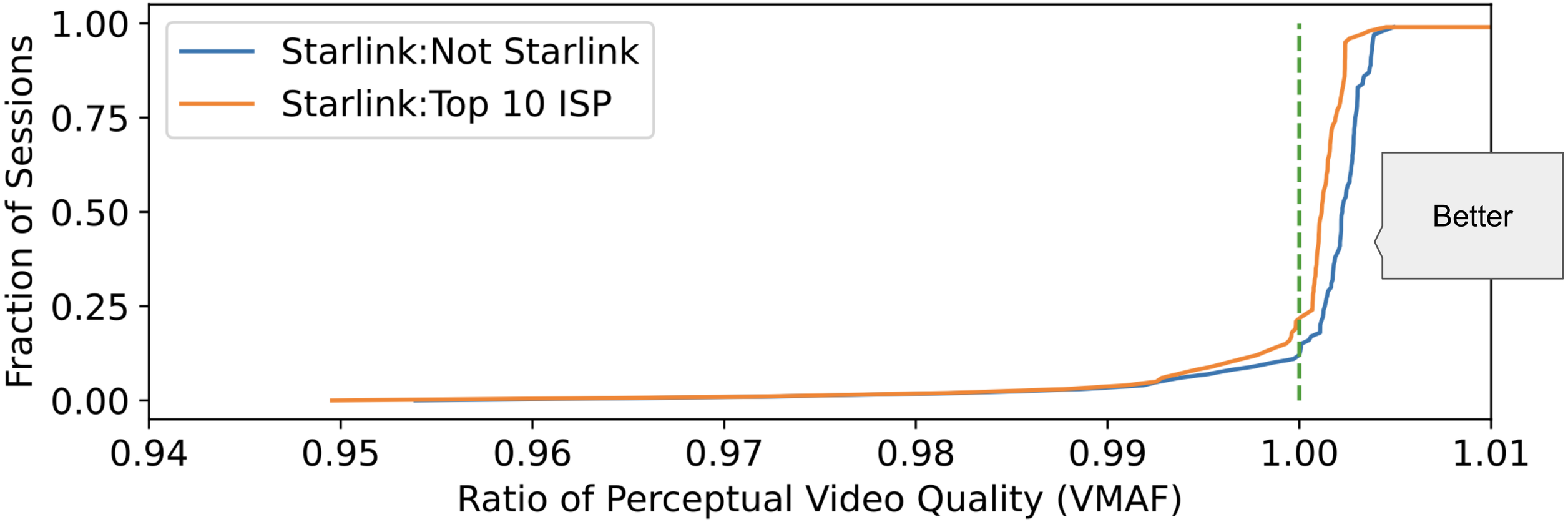
Perceptual Video Quality

Quantifying Perceptual Video Quality Using Video Multi-Method Assessment Fusion (VMAF)

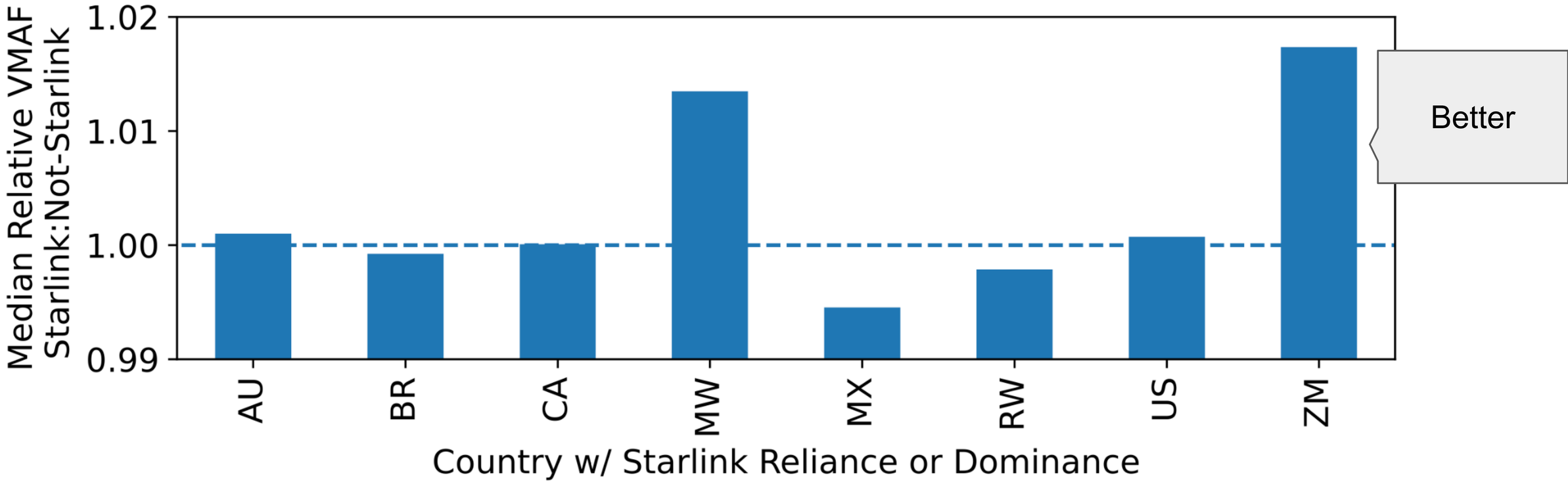
- For each (video, bitrate), predicts how humans will perceive its quality



Starlink users often experience better perceptual video quality



Malawi and Zambia experience better perceptual quality than local alternatives



Bitrate Switches

Video streaming over Starlink suffers from increased bitrate switches



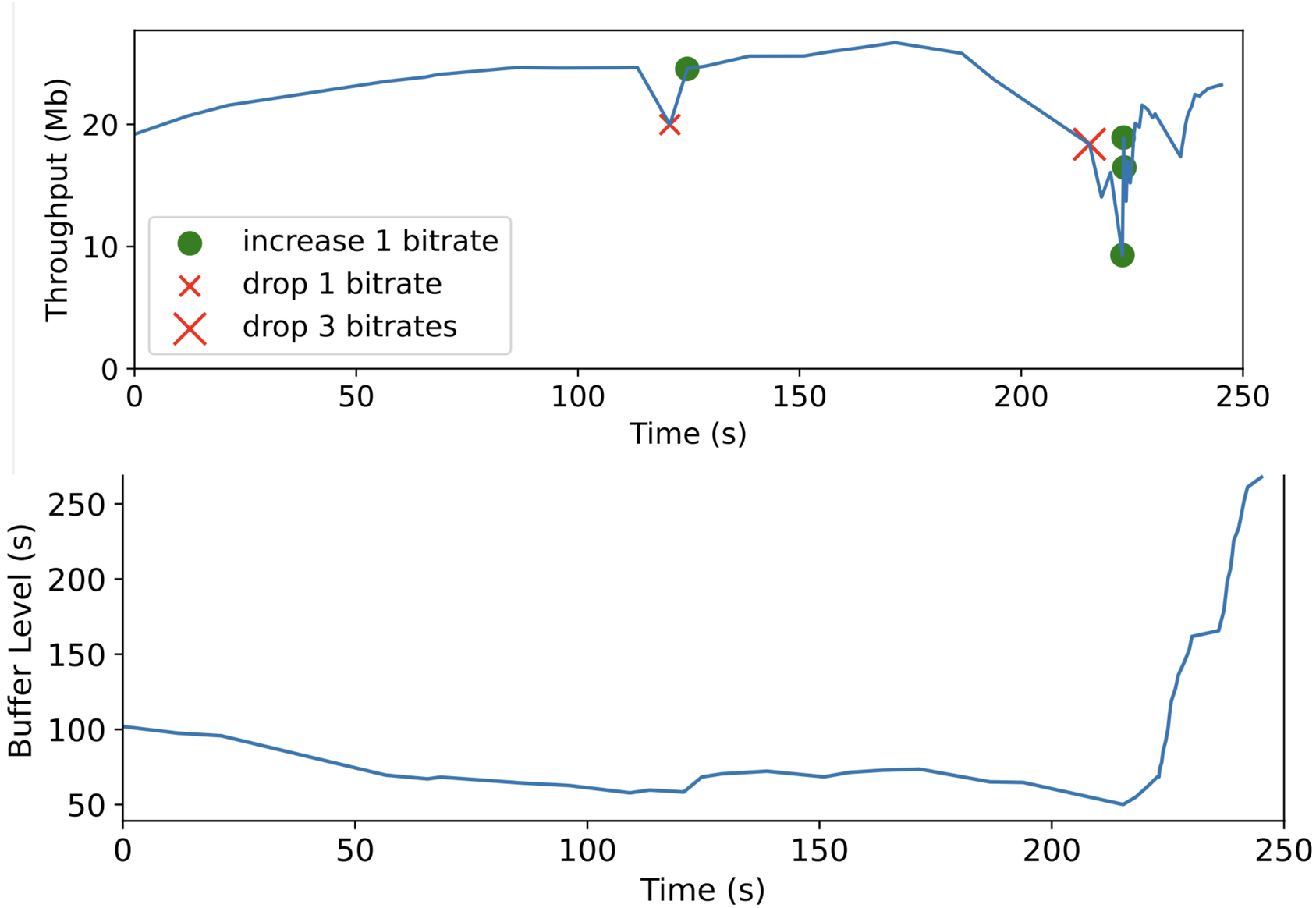
r/Starlink • 2 yr. ago
r00x

Anyone else struggle with poor quality on streaming services (Netflix/Amazon Prime mostly)?

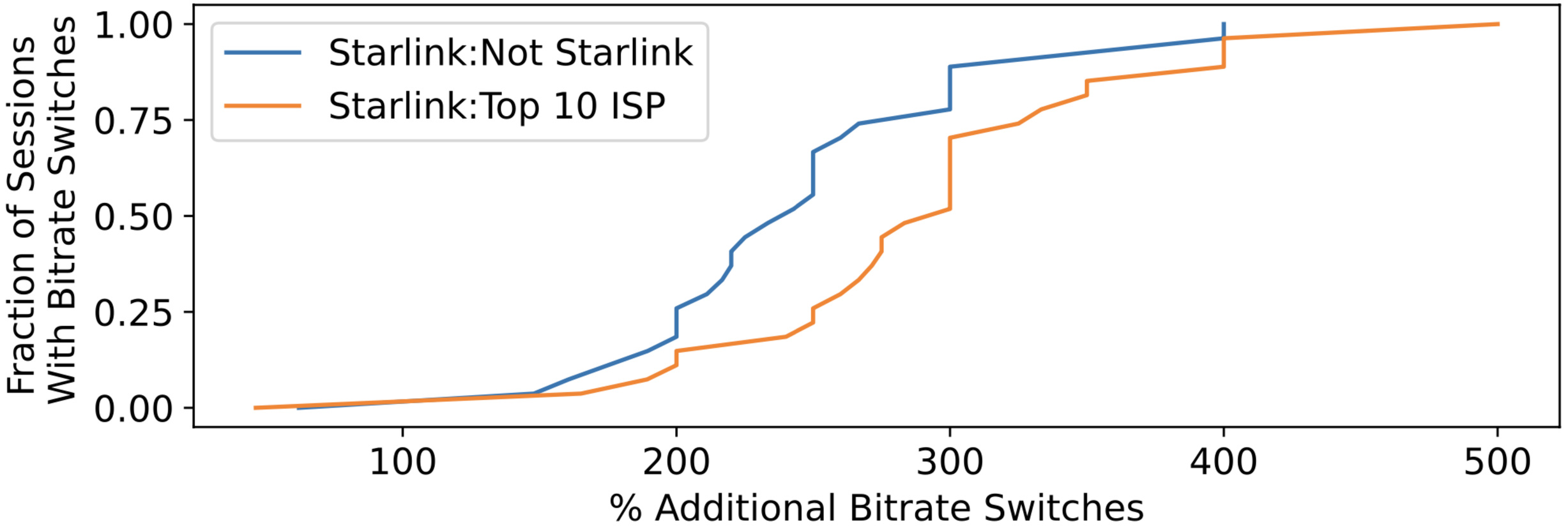
💻 Troubleshooting

Title. I've found Netflix and Amazon Prime are the worst offenders; regularly dropping stream quality from 4K UHD down to lower resolutions, all the way to unwatchable potato quality sometimes (Prime especially), despite the

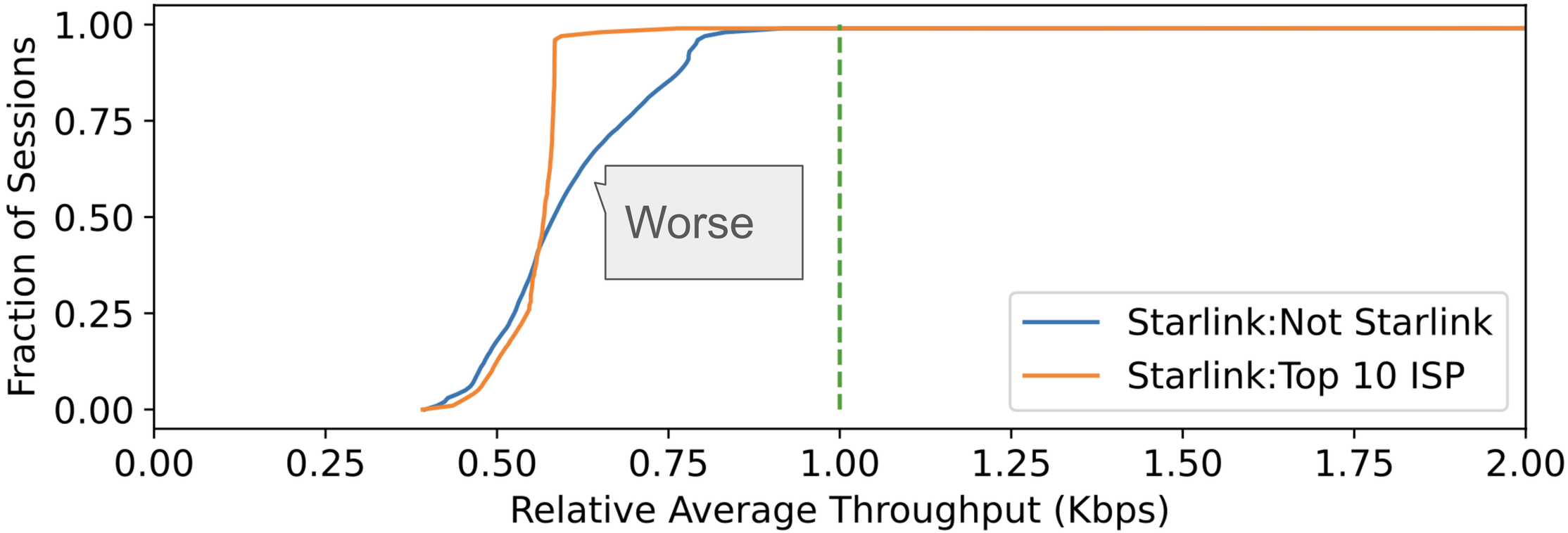
Video streaming over Starlink suffers from increased bitrate switches



Video streaming over Starlink suffers from increased bitrate switches



Starlink’s reduced throughput likely contributes to bitrate switches



Network Rebuffers

Africa and Latin America are more likely to experience a rebuffer over Starlink

Region	Likelihood of >= 1 Rebuffer Relative to US
Canada	0.66x
Asia Pacific	0.9x
United States	1x
Europe	1.07x
Africa	2.7x
Latin America	3.7x

Africa and Latin America are more likely to experience a rebuffer over Starlink

Region	Likelihood of >= 1 Rebuffer Relative to US
Canada	0.66x
Asia Pacific	0.9x
United States	1x
Europe	1.07x
Africa	2.7x
Latin America	3.7x

Starlink sessions spend 50% less time in reduced bitrate state → not reacting quickly enough → buffer not filled enough

Improving Bitrate Switches Using Congestion Control

See paper





Improving Network Rebuffers Using Adaptive Bitrate



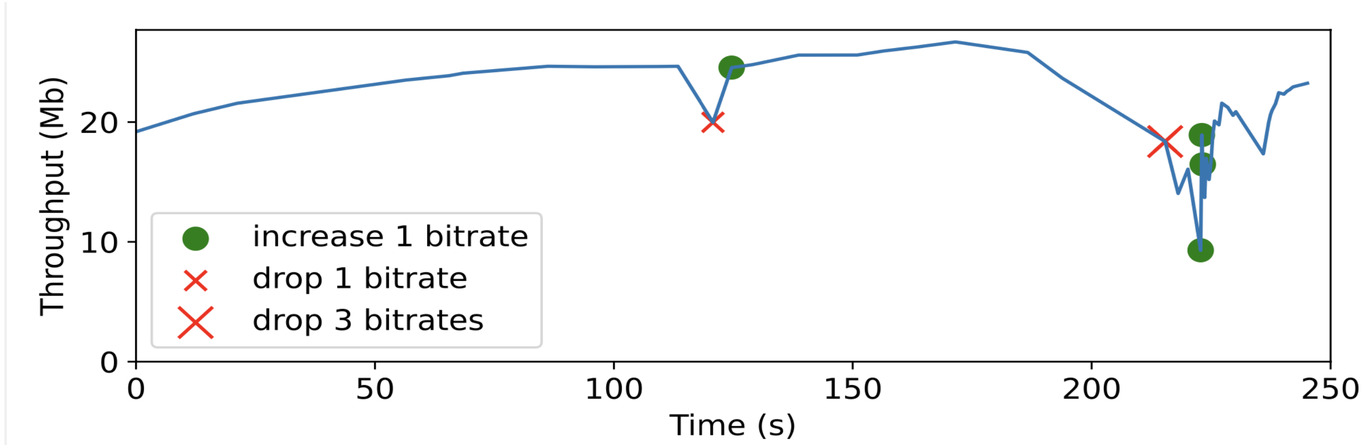


Hypothesis

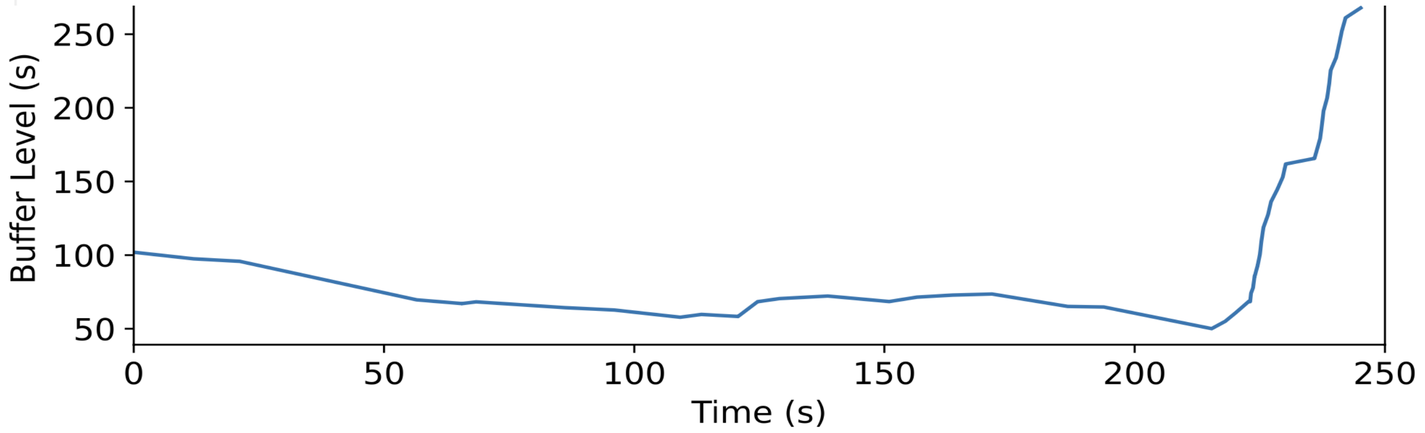
Fewer Network Rebuffers <- “Better” ABR <- Better ABR Parameters for Starlink

Adaptive Bitrate Design Parameters Today

- Throughput Discount
- Buffer Discount
- Throughput Smoothing



Imagine we want to increase buffer level..to avoid rebuffer…

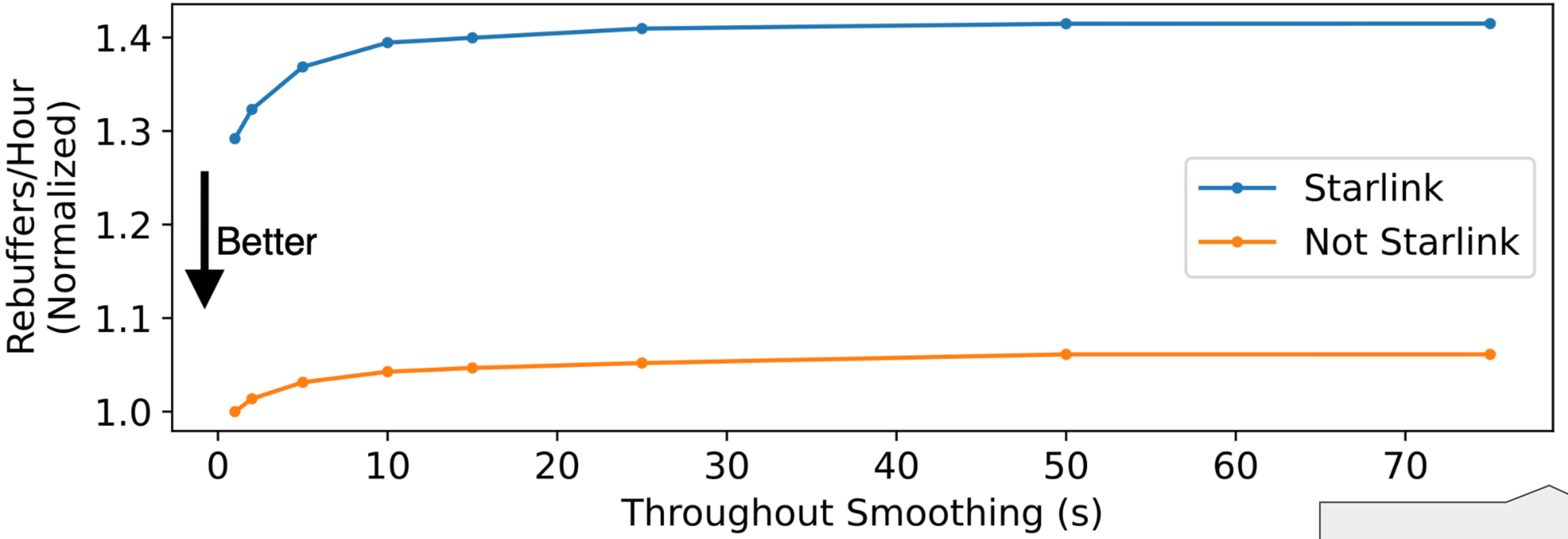


Simulation Methodology

For 500K Starlink and 500K non-Starlink sessions:

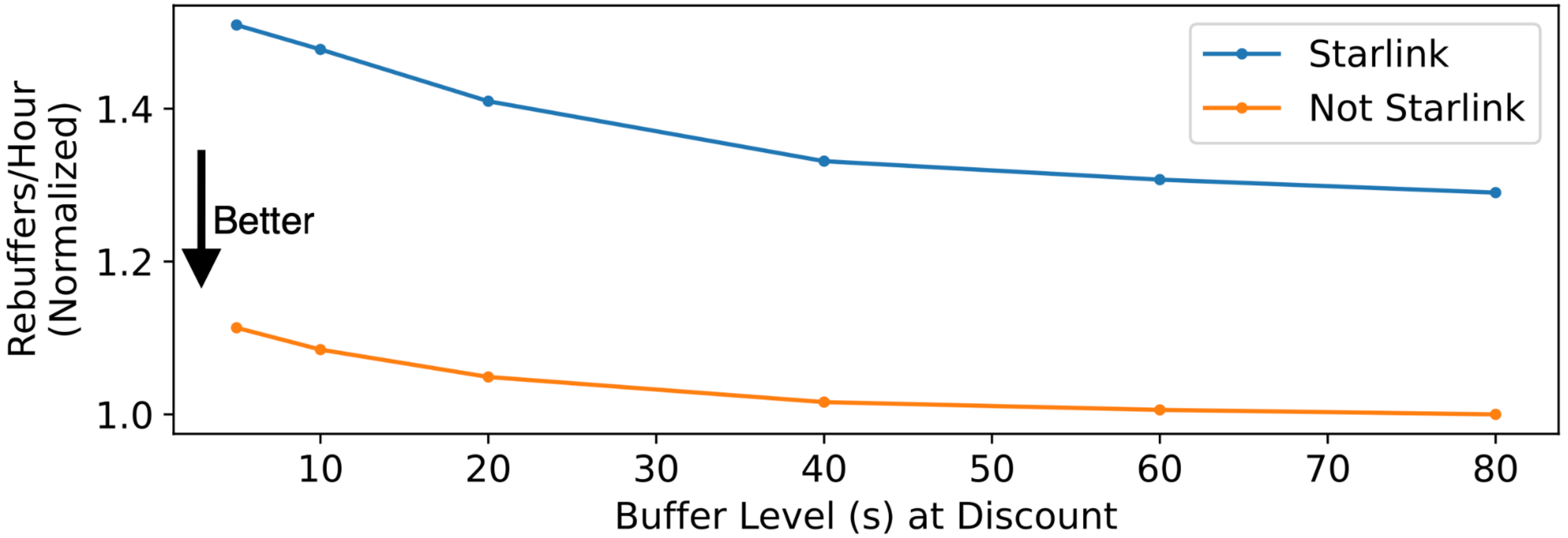
- Vary one of the following
  - Throughput Discount
  - Buffer Level for Discount
  - Throughput Smoothing

Throughput smoothing does not allow Starlink to catch up to other rebuffer rates

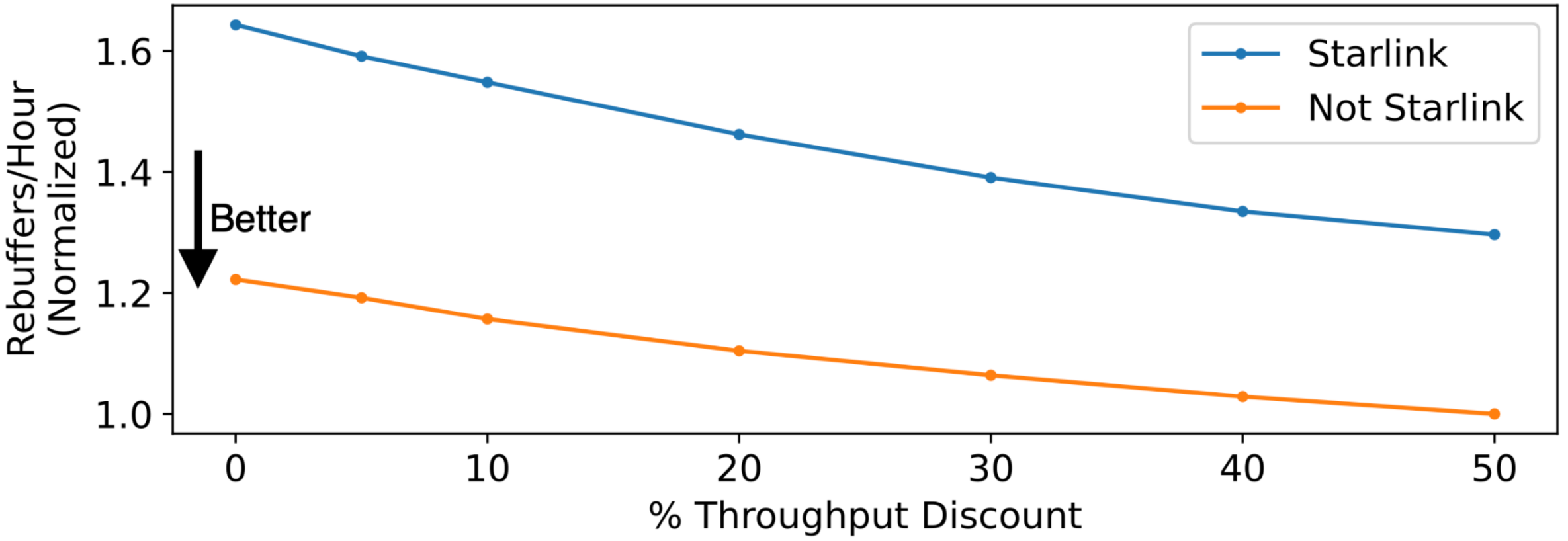


Verified with A/B Test

Buffer discount does not allow Starlink to catch up to other rebuffer rates



Throughput discount does not allow Starlink to catch up to other rebuffer rates



Hypothesis

Fewer Network Rebuffers <- “Better” ABR <- Better ABR Parameters for Starlink





Hypothesis

Fewer Network Rebuffers <- “Better” ABR <- Better ABR Parameters for Starlink

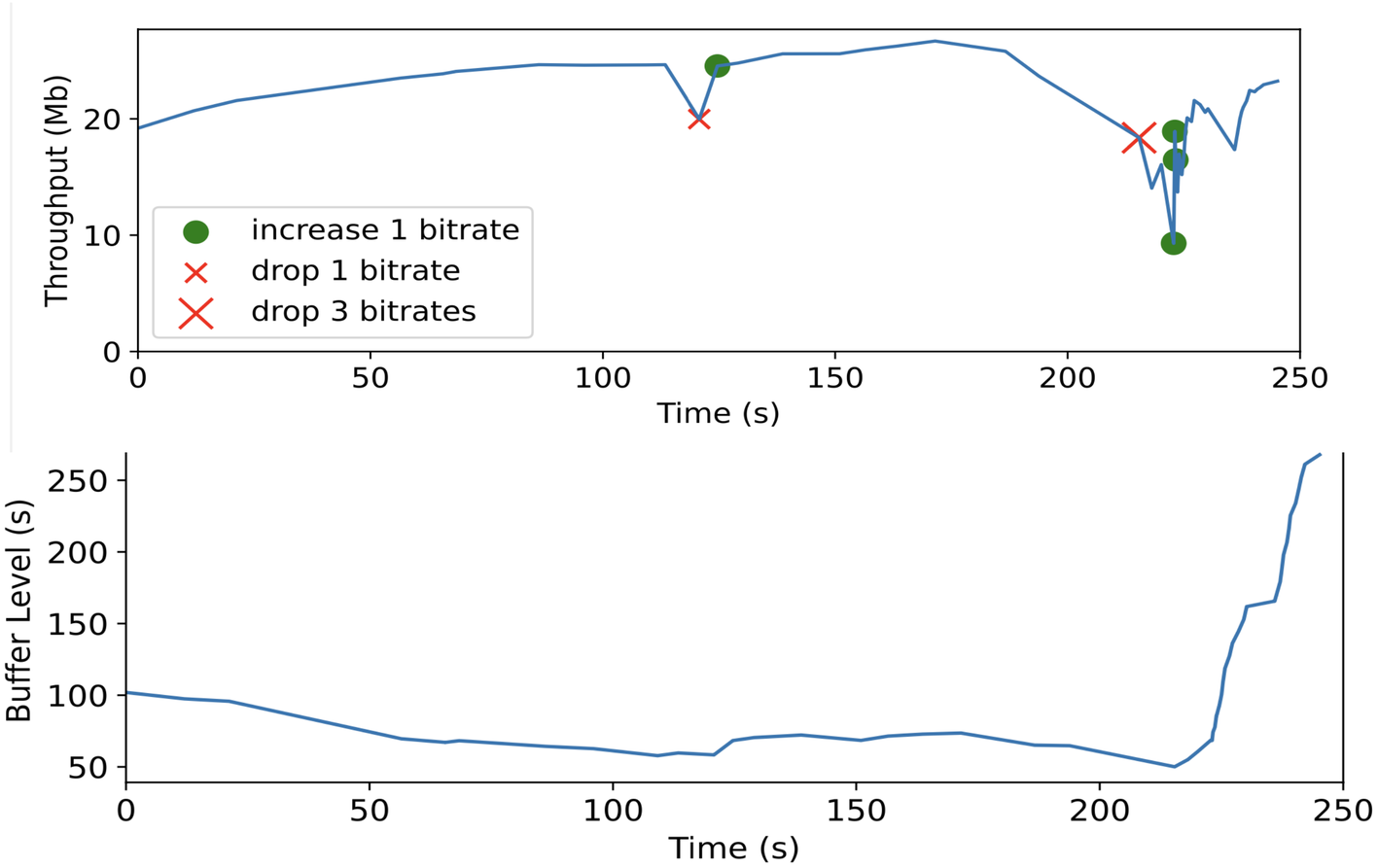




Starlink can never catch up to non-Starlink rebuffer rates



Future of ABR: account for throughput variation, while balancing bitrate switches and rebuffers



Summary

- Video streaming over LEO is rapidly rising in popularity
- Overall quality of experience often better than regional alternatives
- Should reduce rebuffers and bitrate switches for video streaming over LEO
- LEO presents interesting applications for adaptive bitrate streaming, and networking more broadly

Questions?

lizhikev@ucla.edu